

DIPOLE TYPE BY LENGTH

Check first <i>compute $\lambda = c/f$, then l/λ</i>		
l/λ	Type	Use
$< 1/50$	Hertzian (short)	Hertzian formula
$= 1/4$	Quarter-wave	Arbitrary formula
$= 1/2$	Half-wave	$D = 1.64$, simple
other	Arbitrary	Arbitrary formula

If $l/\lambda < 1/50$ stop and use Hertzian only — don't subst. small l into arbitrary formula.

HERTZIAN DIPOLE — $S(R, \theta)$

Power density <i>$l/\lambda < 1/50$; far field</i>
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32 \pi^2 R^2} \sin^2 \theta \text{ (W/m}^2\text{)}$
$k = 2\pi/\lambda \text{ rad/m; } I_0 \text{ peak current (A); } l \text{ length (m); } R \text{ obs. dist (m).}$
Max direction $\theta = \pi/2 \text{ (broadside)}$
Total radiated power <i>integrate over $4\pi \text{ sr}$</i>
$P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$
Directivity <i>Hertzian, exact</i>
$D_{\text{Hertz}} = 1.5 \text{ (i.e. 1.76 dB)}$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at distance R <i>any l</i>
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
Normalized pattern $F(\theta) = S(\theta)/S_{\text{max}}$, <i>dimensionless</i>
Half-wave $l = \lambda/2 \quad \pi l/\lambda = \pi/2$
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}, \quad D = 1.64, \quad R_{\text{rad}} = 73 \Omega$
Quarter-wave $l = \lambda/4 \quad \pi l/\lambda = \pi/4, \cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2} \right)^2 \approx 0.0858 S_0$

BEAM PARAMETERS — HPBW, Ω_p, D

1. Normalize
- $F(\theta) = S(\theta)/S_{\text{max}}$
2. Half-power beamwidth (HPBW) *solve $F(\theta_{1/2}) = 0.5$*
- $\beta = 2\theta_{1/2} \text{ (symmetric beams)}$
- example $F(\theta) = e^{-20\theta^2} \Rightarrow \theta_{1/2} = \sqrt{-\ln 0.5/20} = 0.186 \text{ rad} \Rightarrow \beta = 21.31^\circ$
3. Pattern solid angle *(sr)*
- $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$
- If F has no ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$
4. Directivity
- $D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \quad D_{\text{dB}} = 10 \log_{10} D$
5. Gain $\xi = \text{radiation efficiency, } 0 < \xi \leq 1$
- $G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$
- For lossless / ideal antennas: $G = D$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} \text{ (}\Omega\text{)}$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

EFFECTIVE AREA A_e

Definition m^2 , <i>antenna's receiving cross section</i>
$A_e = \frac{\lambda^2 D}{4\pi}$
Physical cross section <i>half-wave dipole, wire diameter d</i>
$A_p = l d = \frac{\lambda}{2} d$
Ratio $A_e \gg A_p$ <i>for thin wires (good!)</i>
Example: $f = 100 \text{ MHz}, d = 2 \text{ cm} \Rightarrow A_e/A_p \approx 39$.

POWER FLOW IDENTITIES

$P_{\text{rad}} = S_{\text{max}} \Omega_p R^2 = \frac{4\pi R^2 S_{\text{max}}}{D}$
$S_{\text{max}} = \frac{P_{\text{rad}} D}{4\pi R^2} \text{ (W/m}^2\text{)}$
For isotropic radiator: $S = P_{\text{rad}}/(4\pi R^2)$.

FRIIS TRANSMISSION FORMULA

Far-field link budget <i>P_t tx, P_r rx, dist R</i>
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$
G_t, G_r linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.
Solve for R
$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$
Solve for G_r given A_e
$G_r = \frac{4\pi A_{e,r}}{\lambda^2} \Rightarrow P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$
Use linear G , not dB. Convert first: $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$.
Free-space path loss $L_{\text{fs}} = (4\pi R/\lambda)^2$ so $P_r = G_t G_r P_t / L_{\text{fs}}$.

dB $\leftrightarrow$ LINEAR

Power gain <i>factor $\rightarrow$ dB</i>
$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$
dB linear dB linear
0 1 10 10
3 2 13 20
6 4 20 100
7 5 30 1000
−3 0.5 −10 0.1
For E or amplitude use $20 \log$ not $10 \log$.

RADAR EQUATION (BONUS)

Monostatic radar <i>$tx = rx$, target cross section $\sigma_t \text{ (m}^2\text{)}$</i>
$P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}$
Doppler shift <i>closing velocity v_r</i>
$f_d = \frac{2v_r}{\lambda} = \frac{2v_r f}{c} \text{ (Hz)}$
Factor of 2 from round-trip path; v_r is the radial component toward radar.

LECTURE-11 BOUNDARY $\leftrightarrow$ T-LINE

EM	T-line	Match
$\tilde{E}_i$	$\tilde{V}_1^+$	forward wave
$\tilde{E}_r$	$\tilde{V}_1^-$	reverse wave
η_i	Z_{0i}	line impedance
Γ	Γ	same formula
k_i	β_i	wavenumber

Bouncing-wave analogy: every t-line formula in Ch. 2 transfers verbatim with the substitutions above.

ANTENNA INTEGRATION TIPS

Many radiation integrals (e.g. Ω_p for $e^{-20\theta^2} \sin \theta$) have no closed form.
Use Wolfram / numerical integration. State this in the answer.
Common shortcut: if $F(\theta)$ is narrow ($\theta_{1/2} \ll 1$), approximate
$\Omega_p \approx \theta_{1/2}^2 \pi / (4 \ln 2) \text{ (Gaussian beam)}$

DIPOLE

l — antenna length (m)
$\lambda = c/f$ — wavelength (m)
I_0 — peak current (A)
$\eta_0 = 120\pi \approx 377 \Omega$
$k = 2\pi/\lambda$ — wavenumber (rad/m)
R_{rad} — radiation res. (Ω)
R — obs. dist (m)

RADIATION

$S(R, \theta)$ — power density (W/m^2)
S_{max} — peak power density
$F(\theta) = S/S_{\text{max}}$ — normalized
Ω_p — pattern solid angle (sr)
β — HPBW (rad or deg)
D — directivity (unitless)
P_{rad} — total rad. power (W)

ANTENNA

$A_e = \lambda^2 D / (4\pi)$ — effective area
A_p — physical cross sec. (m^2)
ξ — rad. efficiency ($0 < \xi \leq 1$)
$G = \xi D$ — gain (unitless)
$G_{\text{dB}} = 10 \log G$
Half-wave: $D = 1.64, R_{\text{rad}} = 73 \Omega$
Hertzian: $D = 1.5$

FRIIS / LINK

P_t — tx power (W)
P_r — rx power (W)
G_t, G_r — linear gains
R — range (m)
$L_{\text{fs}} = (4\pi R/\lambda)^2$ — path loss
3 dB $\rightarrow$ 2; 10 dB $\rightarrow$ 10
$G_{\text{lin}} = 10^{G_{\text{dB}}/10}$

RADAR

σ_t — RCS (m^2)
f_d — Doppler shift (Hz)
v_r — radial vel. (m/s)
$P_r \propto 1/R^4$ (monostatic)
$f_d = 2v_r/\lambda$ (round trip)
$c = 3 \times 10^8 \text{ m/s}$
$d > 0$: target approaching